

2a	total volume of molecules is negligible compared with volume occupied by the gas	B1
2b	$pV = NkT$ $4.60 \times 10^5 \times 2.40 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times (273 + 23)$	C1

	$N = 2.7 \times 10^{24}$ volume of one molecule = $(4/3)\pi r^3 (= 1.41 \times 10^{-29} \text{ m}^3)$ volume of all molecules = $2.7 \times 10^{24} \times 1.41 \times 10^{-29}$ $= 4 \times 10^{-5} \text{ m}^3$ (1 s.f.; as this is an estimation question the uncertainty of final answer cannot be more precise than that of the given data.) or volume of one molecule = $d^3 (= 2.7 \times 10^{-29} \text{ m}^3)$ volume of all molecules = $2.7 \times 10^{-29} \times 2.7 \times 10^{24}$ $= 7 \times 10^{-5} \text{ m}^3$ (1 s.f.; as this is an estimation question the uncertainty of final answer cannot be more precise than that of the given data.)	C1 A1
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2c	Since volume of all atoms ($4 \times 10^{-5} \text{ m}^3$ or $7 \times 10^{-5} \text{ m}^3$) is <i>3 orders of magnitude</i> less than volume occupied by the gas ($2.4 \times 10^{-2} \text{ m}^3$), so assumption in (a) is justified.	B1
2d	<u>work done on gas (P → Q): 0</u> <u>increase in internal energy (P → Q): (+)97.0 J</u> <u>increase in internal energy (Q → R): −42.5 J</u> <u>work done on gas (R→P)</u> $W = p \Delta V = 2.10 \times 10^5 \times (1125 - 950) \times 10^{-6} = 36.8 \text{ J}$ <u>increase in internal energy (R→P)</u> since total change in internal energy is zero, $97 + (-42.5) + \Delta U = 0 \rightarrow \Delta U = -54.5 \text{ J}$ <u>thermal energy supplied (R→P)</u> $Q = \Delta U - W = -54.5 - 36.8 = -91.3 \text{ J}$	A1 A1 A1 A1 A1

3a	amplitude = 0.020 m	E	C1
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